The Dependency Ordered Formal Exposition of PDEs

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Abstract. A step-indexed, dependency-ordered presentation of Appendices A–E of Evans' *Partial Differential Equations*. The entries are arranged in ten levels of a topological ordering: every result appears only after all results it depends on, with circular dependencies broken at the point where the simpler formulation is introduced first.

Level 0 — Primitive Axioms and Elementary Definitions

1	ax : R — comm : $\forall (x : \mathbb{R})(y : \mathbb{R}), \mathrm{add}(x, y) = \mathrm{add}(y, x)$	Additive Commutativity Axiom
2	ax : R — assoc : $\forall (x : \mathbb{R})(y : \mathbb{R})(z : \mathbb{R}), \mathrm{add}(\mathrm{add}(x, y), z) = \mathrm{add}(x, \mathrm{add}(y, z))$	Additive Associativity Axiom
3	ax : R — order : $\forall (x : \mathbb{R})(y : \mathbb{R})(z : \mathbb{R}), \mathrm{le}(x, y) \wedge \mathrm{le}(y, z) \Rightarrow \mathrm{le}(x, z)$	Transitivity of Order Axiom
4	ax : R — complete : $\forall (a : \mathbb{N} \to \mathbb{R}), \mathrm{Cauchy}(a) \Rightarrow \exists (L : \mathbb{R}), \lim_{n\to\infty} a(n) = L$	Completeness Axiom

5	Def : M : $\mathrm{M}^{m\times n} := \mathrm{space}(\mathbb{R}^{m\times n})$	Definition of Matrix Space
6	Def : S : $\mathrm{S}^n := \mathrm{space}(\mathbb{R}^{n\times n}) \wedge \mathrm{sym}(\mathrm{S}^n)$	Definition of Symmetric Matrix Space
7	Def : tr : $\forall (A \in \mathrm{M}^{n\times n}), \mathrm{tr}(A) = \sum_{i=1}^{n} A_{ii}$	Definition of Trace
8	Def : diag : $\mathrm{diag}(d_1, \ldots, d_n)_{ij} := d_i \delta_{ij}$	Definition of Diagonal Matrix

9	Def : colon : $\forall (A, B \in \mathrm{M}^{m\times n}), \mathrm{colon}(A, B) = \sum_{i,j} A_{ij} B_{ij}$	Definition of Matrix Inner Product
10	Def : quad : $\forall (A \in \mathrm{S}^n)(x \in \mathbb{R}^n), \mathrm{quad}(A, x) = \sum_{i,j} A_{ij} x_i x_j$	Definition of Quadratic Form
11	Def : sphere : $S^{n-1} := \partial B(0, 1) \subset \mathbb{R}^n$	Definition of Unit Sphere
12	Def : halfspace : $\mathbb{R}^n_+ := \{x \in \mathbb{R}^n \mid x_n > 0\}$	Definition of Upper Half-Space

13	Def : compactcontained : $\forall (U, V \subset \mathbb{R}^n), \mathrm{ccontained}(V, U) :\Leftrightarrow V \subset \overline{V} \subset U \wedge \mathrm{compact}(\overline{V})$	Definition of Compact Containment
14	Def : UT : $U_T := U \times (0, T]$	Definition of Parabolic Cylinder
15	Def : alpha : $\mathrm{alpha}(n) := \lvert B(0, 1)\rvert = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$	Definition of Unit Ball Volume
16	Def : spt : $\mathrm{spt}(u) := \overline{\{x \mid u(x) \neq 0\}}$	Definition of Support

17	Def : posneg : $\mathrm{pos}(u) = \max(u, 0), \mathrm{neg}(u) = -\min(u, 0)$	Definition of Positive and Negative Parts
18	Def : sgn : $\mathrm{sgn}(x) := 1_{\{x>0\}} - 1_{\{x<0\}}$	Definition of Sign Function
19	Def : char : $\mathrm{char}(E, x) := 1_{\{x\in E\}}$	Definition of Characteristic Function
20	Def : Lip : $\mathrm{Lip}(u) := \sup_{x\neq y} \frac{\lvert u(x)-u(y)\rvert}{\lvert x-y\rvert}$	Definition of Lipschitz Constant

21	Def : conv : $\mathrm{conv}(f, g)(x) := \int f(x - y) g(y)\, dy$	Definition of Convolution
22	Def : multi : $\forall (\alpha \in \mathbb{N}^n), \mathrm{order}(\alpha) := \alpha_1 + \cdots + \alpha_n$	Definition of Multi-Index Order
23	Def : Dalpha : $\mathrm{Da}(\alpha, u) := \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} u$	Definition of Multi-Index Derivative
24	Def : grad : $\mathrm{grad}(u) := (u_{x_1}, \ldots, u_{x_n})$	Definition of Gradient

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25	Def : bigO : bigO$(f, g, x_0) :\Leftrightarrow \exists C, \forall x$ near$(x, x_0), \lvert f(x)\rvert \leq C\lvert g(x)\rvert$	Definition of Big-O Notation
26	Def : littleo : littleo$(f, g, x_0) :\Leftrightarrow \lim_{x \to x_0} \lvert f(x)\rvert/\lvert g(x)\rvert = 0$	Definition of Little-o Notation
27	rem : grad : $Du \not\equiv \nabla u \wedge D^2 u =$ Hessian $\wedge \nabla^2 u \neq$ Hessian	Remark on Gradient Notation
28	rem : Omega : $U \not\equiv \Omega \wedge \Omega :=$ prob $-$ space	Remark on Domain Notation

29	Def : moll : moll$(\eta) :\Leftrightarrow \eta(x) = C \exp(\frac{1}{\lvert x\rvert^2 - 1}) 1_{\{\lvert x\rvert < 1\}} \wedge \int \eta \, dx = 1$	Definition of Mollifier
30	Def : Norm : Norm$(\lVert \cdot \rVert) :\Leftrightarrow \forall u, v, \lVert u + v\rVert \leq \lVert u\rVert + \lVert v\rVert \wedge \forall \lambda, \lVert \lambda u\rVert = \lvert\lambda\rvert\lVert u\rVert \wedge \lVert u\rVert = 0 \Leftrightarrow u = 0$	Definition of Norm
31	Def : nu : UnitNormal$(\nu, \partial U) :\Leftrightarrow \nu = (\nu^1, \ldots, \nu^n) \wedge \lvert\nu\rvert = 1$	Definition of Unit Normal
32	Def : funcCk : $C^k(U) :=$ {$u \mid u$ k $-$ times $-$ contdiff}	Definition of Ck Functions

33	FTC : $\forall (f : [a, b] \to \mathbb{R}), f \in C^1 \Rightarrow \int_a^b f'(t)\, dt = f(b) - f(a)$	Fundamental Theorem of Calculus
34	Taylor : $\forall (f : \mathbb{R} \to \mathbb{R}), f \in C^2 \wedge x, z \in \mathbb{R} \Rightarrow \exists \xi \in [x, z], f(x) = f(z) + f'(z)(x - z) + \frac{1}{2} f''^2$	Taylor's Theorem with Remainder
35	ProductRule : $\forall u, v \in C^1,\ (uv)' = u'v + uv'$	Product Rule for Derivatives
36	ChainRule : $\forall (f, g) \in C^1,\ (f \circ g)' = f'(g)\, g'$	Chain Rule for Derivatives

37	JacobiFormula : $\forall A(t) \in C^1,\ \frac{d}{dt} \det A = \det A \cdot \mathrm{tr}(A^{-1} A')$	Jacobi's Formula

Level 1

38	Def : normM : $\forall (A \in \mathrm{M}^{m \times n})$, normM$(A) =$ (colon$(A, A))^{1/2}$	Definition of Matrix Norm
39	Def : ge : $\forall (A, B \in \mathrm{S}^n)$, ge$(A, B) :\Leftrightarrow$ nonnegDef(sub(A, B))	Definition of Matrix Order
40	Def : parab : $\Gamma_T := \overline{U}_T \setminus U_T$	Definition of Parabolic Boundary
41	Def : avg $-$ ball : avgB$(f, x, r) :=$ $\frac{1}{\text{alpha}(n) r^n} \int_{B(x,r)} f \, dy$	Definition of Ball Average

42	Def : avg $-$ sph : avgS$(f, x, r) :=$ $\frac{1}{n \, \text{alpha}(n) r^{n-1}} \int_{\partial B(x,r)} f \, dS$	Definition of Sphere Average
43	Def : avg $-$ set : avgE$(f, E) :=$ $\frac{1}{\mu(E)} \int_E f \, d\mu \quad (\mu(E) > 0)$	Definition of Set Average
44	Def : Simple : Simple(s) $:\Leftrightarrow \exists E_i, a_i,$ s $= \sum \chi_{E_i} a_i$	Definition of Simple Function
45	Def : StrongConv : StrongConv$(u, u_0) :\Leftrightarrow \lim_k \lVert u_k - u_0\rVert = 0$	Definition of Strong Convergence

46	Def : perp : perp$(u, v) :\Leftrightarrow (u, v) = 0$	Definition of Orthogonality
47	Def : ONB : $\forall w$, ONB$(w) :\Leftrightarrow \forall k \neq l, (w_k, w_l) = 0 \wedge \forall k, \lVert w_k\rVert = 1$	Definition of Orthonormal Basis
48	Def : Sperp : Sperp$(S) :=$ {$u \mid \forall v \in S, (u, v) = 0$}	Definition of Orthogonal Complement
49	Def : Linear : $\forall A$, Linear$(A) :\Leftrightarrow \forall u, v, \lambda, \mu, A[\lambda u + \mu v] = \lambda Au + \mu Av$	Definition of Linear Operator

50	Def : Bdd : Bdd$(A) :\Leftrightarrow \exists C, \lVert Au\rVert \leq C\lVert u\rVert$	Definition of Bounded Operator
51	Def : ClosedOp : $\forall A$, ClosedOp$(A) :\Leftrightarrow u_k \to u \wedge Au_k \to v \Rightarrow Au = v$	Definition of Closed Operator
52	Def : rho : rho$(A) :=$ {$\eta \mid$ Bijective$(A - \eta I)$}	Definition of Resolvent Set
53	Def : pairing : pairing$(u^*, u) := u^*(u)$	Definition of Dual Pairing

54	Def : Hess : Hess$(u) := (u_{x_i x_j})_{i,j} \in \mathrm{S}^n$	Definition of Hessian
55	Def : Dk : Dk$(k, u) :=$ {$D^\alpha u \mid \lvert\alpha\rvert = k$}	Definition of k-th Order Derivative Set
56	Def : dnu : dnu$(u, \nu) := \nu \cdot$ grad(u)	Definition of Normal Derivative
57	Def : moll $-$ eps : moll $-$ eps$(\eta_\epsilon, \eta) :\Leftrightarrow \eta_\epsilon(x) = \epsilon^{-n} \eta(x/\epsilon)$	Definition of Standard Mollifier

58	Cauchy : $\forall a, b$, Cauchy$(a, b) :\Leftrightarrow ab \leq \frac{a^2}{2} + \frac{b^2}{2}$	Cauchy's Inequality

Level 2

59	Def : InnerProduct : $\forall (\cdot, \cdot)$, InnerProduct$((\cdot, \cdot)) :\Leftrightarrow$ Sym$((\cdot, \cdot)) \wedge$ Bilinear$((\cdot, \cdot)) \wedge \forall u, (u, u) \geq 0 \wedge (u, u) = 0 \Leftrightarrow u = 0$	Definition of Inner Product
60	Def : Lap : Lap$(u) := \sum_i u_{x_i x_i} =$ tr(Hess(u))	Definition of Laplacian
61	Def : Ckbnd : $\forall (U, k)$, Ck $-$ bdy$(U, k) :\Leftrightarrow \forall x^0 \in \partial U\ \exists r > 0, \gamma \in C^k(\mathbb{R}^{n-1}) : U \cap B(x^0, r) =$ {$x_n > \gamma(x_1, \ldots, x_{n-1})$}	Definition of Ck-Boundary
62	Def : LebesguePoint : LebesguePoint$(x_0, f) :\Leftrightarrow \lim_{r \to 0}$ avgB$(\lvert f - f(x_0)\rvert, x_0, r) = 0$	Definition of Lebesgue Point

63	Def : moll $-$ f : moll $-$ f$(f^{\epsilon}, \eta_{\epsilon}, f, U_{\epsilon}) :\Leftrightarrow f^{\epsilon} = \eta_{\epsilon} * f$	Definition of Mollification
64	Def : Dknorm : Dknorm$(k, u) := (\sum_{\lvert\alpha\rvert=k} \lvert D^{\alpha} u\rvert^2)^{1/2}$	Definition of k-th Order Derivative Norm
65	Def : StronglyMeasurable : StronglyMeasurable(f) $:\Leftrightarrow \exists s_k$ Simple$(s_k) \wedge s_k \to$ f a.e.	Definition of Strong Measurability
66	Def : Cauchy : $\forall (u : \mathbb{N} \to H)$, Cauchy$(u) :\Leftrightarrow \forall \epsilon > 0\ \exists N, \forall k, l \geq N, \lVert u_k - u_l \rVert < \epsilon$	Definition of Cauchy Sequence

67	Def : Eigenvalue : Eigenvalue$(A, \eta) :\Leftrightarrow N(A - \eta I) \neq \{0\}$	Definition of Eigenvalue
68	Def : sigma : sigma$(A) := \mathbb{R} \setminus$ rho(A)	Definition of Spectrum
69	Def : Xstar : $X^* := \{\phi : H \to \mathbb{R} \mid$ Linear$(\phi) \wedge$ Bdd$(\phi)\}$	Definition of Dual Space
70	SecondDerivCriterion : $\forall (f : \mathbb{R} \to \mathbb{R}), f \in C^2 \wedge (\forall x \in \mathbb{R}, f''(x) \geq 0) \Rightarrow$ Convex(f)	Second Derivative Convexity Criterion

71	CauchyEps : $\forall a, b, \epsilon > 0$, CauchyEps$(a, b, \epsilon) :\Leftrightarrow ab \leq \epsilon a^2 + \frac{b^2}{4\epsilon}$	Cauchy's Inequality with Epsilon

Level 3

72	Def : SigmaAlgebra : SigmaAlgebra$(\mathcal{M}) :\Leftrightarrow \emptyset, \mathbb{R}^n \in \mathcal{M} \wedge \forall A \in \mathcal{M}, \mathbb{R}^n \setminus A \in \mathcal{M} \wedge \forall \{A_k\} \subset \mathcal{M}, \cup A_k, \cap A_k \in \mathcal{M}$	Definition of Sigma-Algebra
73	Def : Measurable : Measurable$(f) :\Leftrightarrow \forall U \subset \mathbb{R}$ Open$(U), f^{-1}(U) \in \mathcal{M}$	Definition of Measurable Function
74	Def : Summable : Summable$(f) :\Leftrightarrow \int \lvert f\rvert\, dx < \infty$	Definition of Summable Function
75	Def : esssup : esssup$(f) := \inf\{\mu \mid \lvert\{f > \mu\}\rvert = 0\}$	Definition of Essential Supremum

76	Def : Banach : $\forall X$, Banach$(X) :\Leftrightarrow$ Complete$(X) \wedge$ NormedLinear(X)	Definition of Banach Space
77	Def : Hilbert : $\forall H$, Hilbert$(H) :\Leftrightarrow$ Banach$(H) \wedge \exists(\cdot,\cdot)$ InnerProduct$((\cdot,\cdot))$	Definition of Hilbert Space
78	Def : Convex : $\forall (f : \mathbb{R}^n \to \mathbb{R})$, Convex$(f) :\Leftrightarrow \forall x, y\ \forall (\tau : [0,1]), f(\tau x + (1-\tau)y) \leq \tau f(x) + (1-\tau) f(y)$	Definition of Convex Function
79	Def : Reflexive : Reflexive$(X) :\Leftrightarrow (X^*)^* = X$	Definition of Reflexive Space

80	Def : WeakConv : WeakConv$(u, u_0) :\Leftrightarrow \forall u^* \in X^*$, pairing$(u^*, u_k) \to$ pairing(u^*, u_0)	Definition of Weak Convergence
81	Def : adjoint : adjoint$(A, A^*) :\Leftrightarrow \forall u, v, (Au, v) = (u, A^* v)$	Definition of Adjoint Operator
82	Def : WeaklyMeasurable : WeaklyMeasurable(f) $:\Leftrightarrow \forall u^* \in X^*$, Measurable$(t \mapsto$ pairing$(u^*,$ f$(t)))$	Definition of Weak Measurability
83	Def : Separable : $\forall X$, Separable$(X) :\Leftrightarrow \exists D \subset X$, countable$(D) \wedge$ dense(D, X)	Definition of Separable Space

84	expconvex : $\forall (x, y, \tau)$, expconvex$(x, y, \tau) :\Leftrightarrow e^{\tau x + (1-\tau)y} \leq \tau e^x + (1-\tau) e^y$	Convexity of the Exponential
85	Young : $\forall a, b, p, q$, Young$(a, b, p, q) :\Leftrightarrow ab \leq \frac{a^p}{p} + \frac{b^q}{q}$	Young's Inequality
86	CauchySchwarz : $\forall x, y$, CauchySchwarz$(x, y) :\Leftrightarrow \lvert x \cdot y\rvert \leq \lvert x\rvert\lvert y\rvert$	Cauchy–Schwarz Inequality

Level 4

87	Def : AlmostSeparablyValued : AlmostSeparablyValued(f) $:\Leftrightarrow \exists N, \lvert N\rvert = 0 \wedge$ Separable$(\{$f$(t) \mid t \notin N\})$	Definition of Almost Separably Valued
88	Def : Symmetric : Symmetric$(A) :\Leftrightarrow A^* = A$	Definition of Symmetric Operator
89	Def : CompactOp : CompactOp$(K) :\Leftrightarrow \forall \{u_k\}Bdd(u_k) \Rightarrow \{Ku_k\}$precompact	Definition of Compact Operator
90	BanachFP : $\forall T : X \to X$, Contraction$(T) \wedge$ Complete$(X) \Rightarrow \exists! x^*, Tx^* = x^*$	Banach Fixed-Point Lemma

91	SeparatingHyperplane : $\forall C \subseteq \mathbb{R}^n$, Convex$(C) \Rightarrow \forall p \in \partial C\ \exists u \neq 0,\ \forall z \in C,\ u \cdot (z - p) \leq 0$	Supporting Hyperplane Lemma
92	OpenMap : $\forall T \in \mathcal{L}(X, Y),$ Surjective$(T) \Rightarrow T$ open	Open Mapping Lemma
93	HardyLittlewood : $\forall f \in L^1_{\mathrm{loc}},$ avgB$(\lvert f - f(x)\rvert, x, r) \to 0$ a.e.	Hardy–Littlewood Lemma
94	BanachAlaoglu : $\forall X,$ Reflexive$(X) \Rightarrow \overline{B}(0,1)$ weakly $-$ compact	Banach–Alaoglu Lemma

95	CauchySchwarzA : $\forall A, x, y$, CauchySchwarzA$(A, x, y) :\Leftrightarrow A \geq 0 \wedge A^T = A \Rightarrow \lvert \sum a_{ij} x_i y_j\rvert \leq (\sum a_{ij} x_i x_j)^{1/2} (\sum a_{ij} y_i y_j)^{1/2}$	Cauchy–Schwarz Inequality for Positive Matrices

Level 5

96	YoungEps : $\forall a, b, \epsilon, p, q$, YoungEps$(a, b, \epsilon, p, q) :\Leftrightarrow ab \leq \epsilon a^p + C(\epsilon) b^q$	Young's Inequality with Epsilon
97	Riesz : $\forall u^* \in H^*\ \exists! u \in H, \forall v$, pairing$(u^*, v) = (u, v)$	Riesz Representation Theorem
98	BoundsSpectrum : $\forall S, m, M$, BoundsSpectrum$(S, m, M) :\Leftrightarrow$ sigma$(S) \subset [m, M] \wedge m, M \in$ sigma(S)	Bounds on the Spectrum
99	InverseFT : $\forall$ f, $Jf(x_0) \neq 0 \Rightarrow \exists V, W$, Bijective$(\mathrm{f} : V \to W) \wedge \mathrm{f}^{-1} \in C^1$	Inverse Function Theorem

100	ClosedGraph : $\forall A$, ClosedOp$(A) \wedge$ Linear$(A) \Rightarrow$ Bdd(A)	Closed Graph Theorem
101	GaussGreen : $\forall u, \mathrm{u}, \int_U u_{x_i} = \int_{\partial U} u\nu^i \, dS \wedge \int_U \mathrm{div(u)} = \int_{\partial U} \mathrm{u} \cdot \nu \, dS$	Gauss–Green Theorem
102	Coarea : $\forall u, f, \int_{\mathbb{R}} n \; f \lvert \mathrm{grad}(u) \rvert = \int_{-\infty}^{\infty} \left(\int_{\{u=r\}} f \, dS \right) dr$	Coarea Formula
103	Polar : $\forall f, \int_{\mathbb{R}} n \; f = \int_0^{\infty} \left(\int_{\partial B(x_0, r)} f \, dS \right) dr$	Polar Coordinates Formula

104	MonotoneConv : $\forall f_k, 0 \leq f_1 \leq f_2 \leq \cdots \Rightarrow \int \lim f_k = \lim \int f_k$	Monotone Convergence Theorem
105	Fatou : $\forall f_k$, NonnegMeasurable$(f_k) \Rightarrow \int \liminf f_k \leq \liminf \int f_k$	Fatou's Lemma
106	DominatedConv : $\forall f_k, f, g,$ a.e. $-$ conv$(f_k, f) \wedge \lvert f_k \rvert \leq g$ a.e. $\wedge$ Summable$(g) \Rightarrow \int f_k \to \int f$	Dominated Convergence Theorem
107	WeakCompact : Reflexive$(X) \wedge$ Bdd$(\{u_k\}) \Rightarrow \exists \{u_{k_j}\}, u,$ WeakConv(u_{k_j}, u)	Weak Compactness Theorem

Level 6

108	Holder : $\forall u, v,$ Holder$(u, v) :\Leftrightarrow \int_U \lvert uv \rvert \leq \lVert u \rVert_{L^p} \lVert v \rVert_{L^q}$	Hölder's Inequality
109	IntParts : $\forall u, v, \int_U u_{x_i} v = -\int_U u v_{x_i} + \int_{\partial U} uv\nu^i \, dS$	Integration by Parts
110	MovingRegions : $\forall f, U, \frac{d}{d\tau} \int_{U(\tau)} f = \int_{\partial U(\tau)} f \mathrm{v} \cdot \nu \, dS + \int_{U(\tau)} f_\tau$	Reynolds Transport Theorem
111	LebesgueDiff : $\forall f,$ LocSummable$(f) \Rightarrow$ a.e.$(\mathrm{avgB}(f, x_0, r) \to f(x_0)) \wedge$ a.e.$(\mathrm{avgB}(\lvert f - f(x_0) \rvert, x_0, r) \to 0)$	Lebesgue Differentiation Theorem

112	LebesgueExistence : $\exists \mathcal{M}, \lvert \cdot \rvert,$ SigmaAlgebra$(\mathcal{M}) \wedge$ Open$(U) \wedge$ Closed$(U) \subset \mathcal{M} \wedge \lvert B \rvert =$ vol$(B) \wedge$ countably $-$ additive$(\lvert \cdot \rvert)$	Existence of Lebesgue Measure
113	Egoroff : $\forall f_k, f, A,$ a.e. $-$ on$(f_k, f, A) \wedge \lvert A \rvert < \infty \wedge \epsilon > 0 \Rightarrow \exists E \subset A, \lvert A \setminus E \rvert \leq \epsilon \wedge$ unif $-$ on(f_k, f, E)	Egoroff's Theorem
114	CompactAdjoint : CompactOp$(K) \Rightarrow$ CompactOp(K^*)	Compactness of the Adjoint
115	Pettis : $\forall$ f, StronglyMeasurable(f) $\Leftrightarrow$ WeaklyMeasurable(f) $\wedge$ AlmostSeparablyValued(f)	Pettis's Measurability Theorem

Level 7

116	Minkowski : $\forall u, v,$ Minkowski$(u, v) :\Leftrightarrow \lVert u + v \rVert_{L^p} \leq \lVert u \rVert_{L^p} + \lVert v \rVert_{L^p}$	Minkowski's Inequality
117	GenHolder : $\forall u_k, p_k,$ GenHolder$(u_k, p_k) :\Leftrightarrow \int \lvert u_1 \cdots u_m \rvert \leq \prod_k \lVert u_k \rVert_{L^{p_k}}$	Generalized Hölder's Inequality
118	Interpolation : $\forall u, r, s, t, \theta,$ Interpolation$(u, r, s, t, \theta) :\Leftrightarrow \lVert u \rVert_{L^r} \leq \lVert u \rVert_{L^s}^{\theta} \lVert u \rVert_{L^t}^{1-\theta}$	Interpolation Inequality for Lp Spaces
119	MollifierProps : $\forall f, f^\epsilon, f^\epsilon \in C^\infty(U_\epsilon) \wedge f^\epsilon \to f$ a.e. $\wedge$ $(f \in C(U) \Rightarrow$ unif $-$ on $-$ compacta$(f^\epsilon, f)) \wedge (f \in L^p_{\mathrm{loc}} \Rightarrow f^\epsilon \to f$ in $L^p_{\mathrm{loc}})$	Properties of Mollifiers

120	GreenI : $\forall u, \int_U$ Lap$(u) = \int_{\partial U}$ dnu$(u, \nu) \, dS$	Green's First Identity
121	GreenII : $\forall u, v, \int_U$ grad$(v) \cdot$ grad$(u) = -\int_U u$Lap$(v) + \int_{\partial U} udnu(v, \nu) \, dS$	Green's Second Identity
122	ImplicitFT : $\forall$ f, J_yf$(x_0, y_0) \neq 0 \Rightarrow \exists$ g $\in C^1$, f$(x,$ g$(x)) = z_0$	Implicit Function Theorem
123	Fredholm : CompactOp$(K) \Rightarrow$ $\dim N(I-K) < \infty \wedge$ ClosedOp$(R(I-K))$ $\wedge R(I-K) =$ Sperp$(N(I-K^*))$ $\wedge (N(I-K) = \{0\} \Leftrightarrow R(I-K) = H)$ $\wedge \dim N(I-K) = \dim N(I-K^*)$	Fredholm Alternative

124	Bochner : $\forall$ f, Summable(f) $\Leftrightarrow$ Summable$(t \mapsto \lVert$f$(t) \rVert) \wedge \lVert \int$ f$\rVert \leq \int \lVert$f$\rVert$	Bochner's Theorem
125	ArzelaAscoli : $\forall \{f_k\},$ UnifBdd$(f_k) \wedge$ Equicont$(f_k) \Rightarrow \exists \{f_{k_j}\}, f,$ Continuous$(f) \wedge$ unif $-$ on $-$ compacta(f_{k_j}, f)	Arzelà–Ascoli Theorem
126	SupportingHyperplanes : $\forall f,$ Convex$(f) \Rightarrow \forall x \; \exists r, \forall y, f(y) \geq f(x) + r \cdot (y - x)$	Supporting Hyperplane Theorem
127	LaxMilgram : $\forall a(\cdot, \cdot),$ Coercive$(a) \wedge$ Bdd$(a) \wedge f \in H^* \Rightarrow \exists! u \in H, \forall v, a(u, v) = f(v)$	Lax–Milgram Lemma

Level 8

128	GreenIII : $\forall u, v, \int_U (u$Lap$(v) - v$Lap$(u)) = \int_{\partial U} (udnu(v, \nu) - vdnu(u, \nu)) \, dS$	Green's Third Identity
129	Jensen : $\forall f,$ u, Convex$(f) \Rightarrow f($avgE$($u$, U)) \leq$ avgE$(f \circ$ u$, U)$	Jensen's Inequality
130	SpectrumCompact : $\dim H = \infty \wedge$ CompactOp$(K) \Rightarrow 0 \in$ sigma$(K) \wedge$ sigma$(K) \setminus \{0\} =$ sigma$_p(K) \setminus \{0\} \wedge$ finite $-$ or $-$ tends0(sigma$(K) \setminus \{0\}$)	Spectrum of a Compact Operator

Level 9

131	SpectralThm : Separable$(H) \wedge$ CompactOp$(S) \wedge$ Symmetric$(S) \Rightarrow \exists$ ONB $-$ of $-$ eigenvectors(S)	Spectral Theorem for Compact Symmetric Operators

Proofs

Lemmas

Proof of Lemma 33 (FTC)

142	$f \in C^1([a,b])$	Assumption
143	$F(x) := \int_a^x f'(t)\,dt \wedge F' = f'$ a.e.	Def
144	$\exists C \in \mathbb{R},\ F = f + C$	Eq(143)
145	$F(a) = 0 \Rightarrow C = -f(a) \Rightarrow F(b) = f(b) - f(a)$	Eq(144)

146	$\top$	$\rightarrow$ I(142–145)

Proof of Lemma 34 (Taylor)

134	$f \in C^2(\mathbb{R}) \wedge x, z \in \mathbb{R}$	Assumption
135	$\int_z^x f''(t)(x-t)\,dt = \left[-f'(t)(x-t)\right]_z^x + \int_z^x f'(t)\,dt$	Eq(134)
136	$\int_z^x f''(t)(x-t)\,dt = \int_z^x f'(t)\,dt - f'(z)(x-z)$	Eq(135)
137	$\int_z^x f'(t)\,dt = f(x) - f(z)$	Lem:33(134)

138	$f(x) = f(z) + f'(z)(x-z) + \int_z^x f''(t)(x-t)\,dt$	Eq(136, 137)
139	$\exists \xi \in [x,z], \int_z^x f''(t)(x-t)\,dt = \frac{1}{2} f''^2$	Eq(138)
140	$f(x) = f(z) + f'(z)(x-z) + \frac{1}{2} f''^2$	Eq(138, 139)
141	$\top$	$\rightarrow$ I(134–140)

Proof of Lemma 35 (ProductRule)

147	$u, v \in C^1$	Assumption
148	$(uv)(x+h) - (uv)(x) = u(x+h)v(x+h) - u(x)v(x)$	Def
149	$= [u(x+h) - u(x)]v(x+h) + u(x)[v(x+h) - v(x)]$	Eq(148)
150	$\lim_{h\to 0}(148)/h = u'(x)v(x) + u(x)v'(x)$	Eq(147, 149)

151	$\top$	$\rightarrow$ I(147–150)

Proof of Lemma 36 (ChainRule)

152	$f, g \in C^1$	Assumption
153	$(f \circ g)(x+h) - (f \circ g)(x) = f(g(x+h)) - f(g(x))$	Def
154	$= f'(g(x))[g(x+h) - g(x)] + o(g(x+h) - g(x))$	Eq(153)
155	$\lim_{h\to 0}(153)/h = f'(g(x))\,g'(x)$	Eq(152, 154)

156	$\top$	$\rightarrow$ I(152–155)

Proof of Lemma 37 (JacobiFormula)

157	$A(t) \in C^1 \wedge \det A \neq 0$	Assumption
158	$\det(A + \epsilon B) = \det A + \epsilon\,\mathrm{tr}(\mathrm{adj}(A)B) + O(\epsilon^2)$	Eq
159	$\mathrm{adj}(A) = \det A \cdot A^{-1}$	Def
160	$\frac{d}{dt}\det A = \det A \cdot \mathrm{tr}(A^{-1}A')$	Eq(158, 159)

161	$\top$	$\rightarrow$ I(157–160)

Proof of Lemma 59 (Cauchy)

216	$0 \leq (a-b)^2$	Ax : 3
217	$0 \leq a^2 - 2ab + b^2$	Eq(216)
218	$2ab \leq a^2 + b^2$	addMon(217)
219	Cauchy(a,b)	Eq(218)

220	$\top$	$\rightarrow$ I(216–219)

Proof of Lemma 74 (SecondDerivCriterion)

202	$f \in C^2(\mathbb{R}) \wedge \forall x \in \mathbb{R}, f''(x) \geq 0$	Assumption
203	$x, y \in \mathbb{R} \wedge t \in [0,1] \wedge z := (1-t)x + ty$	Intro
204	$\exists \xi_1,\ f(x) = f(z) + f'(z)(x-z) + \frac{1}{2} f''(\xi_1)(x-z)^2$	Lem:34(202, 203)
205	$f(x) \geq f(z) + f'(z)(x-z)$	$\leq$ E(204, 202)

206	$\exists \xi_2,\ f(y) = f(z) + f'(z)(y-z) + \frac{1}{2} f''(\xi_2)(y-z)^2$	Lem:34(202, 203)
207	$f(y) \geq f(z) + f'(z)(y-z)$	$\leq$ E(206, 202)
208	$(1-t)f(x) + tf(y) \geq f(z)$	addMon(205, 207)
209	Convex(f)	Def:78(203, 208)

210	$\top$	$\rightarrow$ I(202–209)

Proof of Lemma 75 (CauchyEps)

221	$a, b > 0 \wedge \epsilon > 0$	Assumption
222	$ab = ((2\epsilon)^{1/2} a)(b/(2\epsilon)^{1/2})$	Eq
223	CauchyEps(a, b, ϵ)	Lem:58(222)
224	$\top$	$\rightarrow$ I(221–223)

Proof of Lemma 89 (expconvex)

211	$f(t) := e^t \wedge f''(t) = e^t > 0$	Def
212	Convex(f)	Lem:70(211)
213	$f(\tau x + (1-\tau)y) \leq \tau f(x) + (1-\tau) f(y)$	Def:78(212)
214	expconvex(x, y, τ)	Eq(213)

215	$\top$	$\rightarrow$ I(211–214)

Proof of Lemma 90 (Young)

225	$1 < p, q < \infty \wedge p^{-1} + q^{-1} = 1 \wedge a, b > 0$	Assumption
226	Convex(exp)	Lem:84
227	$t := p^{-1} \wedge x := \log a^p \wedge y := \log b^q$	Intro
228	$e^{(1/p)\log a^p + (1/q)\log b^q} \leq \frac{1}{p} e^{\log a^p} + \frac{1}{q} e^{\log b^q}$	Def:78(226, 227)

229	Young(a, b, p, q)	Eq(228)
230	$\top$	$\rightarrow$ I(225–229)

Proof of Lemma 91 (CauchySchwarz)

258	$y = 0 \Rightarrow$ CauchySchwarz(x, y)	Eq
259	$y \neq 0 \Rightarrow 0 \leq \lvert x \pm \epsilon y\rvert^2 = \lvert x\rvert^2 \pm 2\epsilon x \cdot y + \epsilon^2 \lvert y\rvert^2$	Ax : 3
260	$\pm x \cdot y \leq \frac{1}{2\epsilon} \lvert x\rvert^2 + \frac{\epsilon}{2} \lvert y\rvert^2$	addMon(259)
261	$\epsilon = \lvert x\rvert / \lvert y\rvert \Rightarrow \pm x \cdot y \leq \lvert x\rvert \lvert y\rvert$	Eq(260)

262	CauchySchwarz(x, y)	$\leq$ E(258, 261)
263	$\top$	$\rightarrow$ I(258–262)

Proof of Lemma 95 (BanachFP)

177	Contraction$(T) \wedge$ Complete(X)	Assumption
178	$\exists \theta < 1,\ \forall x, y,\ d(Tx, Ty) \leq \theta\, d(x, y)$	Def(177)
179	Sequence$(x_{n+1} := T(x_n))$	Def
180	$d(x_{n+1}, x_n) \leq \theta^n d(x_1, x_0)$	Eq(178, 179)

181	Cauchy(x_n)	$\leq$ E(180)
182	$\exists x^*,\ x_n \to x^*$	$\exists$E(177, 181)
183	$Tx^* = x^* \wedge$ Uniq(x^*)	Eq(179, 182)
184	$\top$	$\to$ I(177–183)

Proof of Lemma 96 (SeparatingHyperplane)

162	$C \subseteq \mathbb{R}^n \wedge$ Convex$(C) \wedge p \in \partial C$	Assumption
163	$\exists p_k \in \mathbb{R}^n \setminus \overline{C},\ p_k \to p$	Def(162)
164	$\exists \pi_k \in \overline{C},\ \|p_k - \pi_k\| =$ dist(p_k, C)	$\exists$I(163)
165	$\forall z \in C,\ (p_k - \pi_k) \cdot (z - \pi_k) \leq 0$	Def:78(162, 164)

166	$u_k := (p_k - \pi_k)/\|p_k - \pi_k\|$	Def
167	Bdd$(u_k) \Rightarrow \exists\{u_{k_j}\},\ u_{k_j} \to u,\ \|u\| = 1$	$\exists$E(166)
168	$\forall z \in C,\ u \cdot (z - p) \leq 0$	Eq(165, 166, 167)
169	$\top$	$\to$ I(162–168)

Proof of Lemma 97 (OpenMap)

185	$T \in \mathcal{L}(X, Y) \wedge$ Surjective(T)	Assumption
186	$X = \cup_{n \in \mathbb{N}} \overline{B}(0, n) \wedge Y = T(X) = \cup_n T(\overline{B}(0, n))$	Eq(185)
187	$\exists n,\ \overline{T(\overline{B}(0, n))}$ has — nonempty — interior	$\exists$E(186)
188	$\exists r > 0,\ \overline{B}_Y(0, r) \subseteq T(\overline{B}(0, n))$	$\exists$I(187)

189	$T(B(0, 1)) \supseteq B_Y(0, r/n)$	Eq(188)
190	T open	Def(189)
191	$\top$	$\to$ I(185–190)

Proof of Lemma 98 (HardyLittlewood)

197	$f \in L^1_{\text{loc}}$	Assumption
198	$M(f)(x) := \sup_{r>0}$ avgB$(\|f\|, x, r)$	Def:41(197)
199	$\|\text{WeakL}^1(M(f))\| \leq C\|f\|_{L^1}$	$\leq$ E(198)
200	avgB$(\|f - f(x)\|, x, r) \to 0$ a.e.	Eq(197, 199)

201	$\top$	$\to$ I(197–200)

Proof of Lemma 99 (BanachAlaoglu)

192	Reflexive(X)	Assumption
193	$\overline{B}(0, 1) \subseteq X = (X^*)^*$	Def:79(192)
194	$\overline{B}(0, 1)$ weak*-compact in $(X^*)^*$	$\exists$E(193)
195	$\overline{B}(0, 1)$ weakly — compactin$X$	Eq(193, 194)

196	$\top$	$\to$ I(192–195)

Proof of Lemma 100 (CauchySchwarzA)

264	$A \geq 0 \wedge A^T = A$	Assumption
265	InnerProduct$((x, y) \mapsto x^T A y)$	Def:59
266	CauchySchwarzA(A, x, y)	Lem:86(265)
267	$\top$	$\rightarrow$ I(264–266)

Proof of Lemma 101 (YoungEps)

231	$a, b > 0 \wedge \epsilon > 0 \wedge 1 < p, q < \infty \wedge p^{-1} + q^{-1} = 1$	Assumption
232	$ab = ((\epsilon p)^{1/p} a)(b/(\epsilon p)^{1/p})$	Eq
233	$ab \leq \epsilon a^p + \frac{b^q}{q(\epsilon p)^{q/p}}$	Lem:85(232)
234	$C(\epsilon) := (\epsilon p)^{-q/p} q^{-1} \Rightarrow$ YoungEps(a, b, ϵ, p, q)	Def(233)

235	$\top$	$\rightarrow$ I(231–234)

Proof of Lemma 103 (BoundsSpectrum)

282	$\eta > M \Rightarrow (\eta u - Su, u) \geq (\eta - M)\|u\|^2$	$\leq$ E
283	Bijective$(\eta I - S)$	Lem:127(282)
284	$\eta < m \Rightarrow (Sv - \eta v, v) \geq (m - \eta)\|v\|^2$	$\leq$ E
285	sigma$(S) \subset [m, M]$	$\forall$I(283, 284)

286	$\|(Mu - Su, v)\| \leq (Mu - Su, u)^{1/2}(Mv - Sv, v)^{1/2}$	Lem:86(285)
287	$\|Mu - Su\| \leq C(Mu - Su, u)^{1/2}$	$\leq$ E(286)
288	$\|u_k\| = 1 \wedge (Su_k, u_k) \to M \Rightarrow \|Mu_k - Su_k\| \to 0$	$\leq$ E(287)
289	$M \in$ rho$(S) \Rightarrow u_k = (MI - S)^{-1}(Mu_k - Su_k) \to 0$	$\perp$I(288)

290	$M \in$ sigma$(S) \wedge m \in$ sigma(S)	$\forall$E(288, 289)
291	$\top$	$\rightarrow$ I(282–290)

Proof of Lemma 132 (LaxMilgram)

170	Coercive$(a) \wedge$ Bdd$(a) \wedge f \in H^*$	Assumption
171	$\forall u\ \exists! Au \in H,\ \forall v,\ (Au, v) = a(u, v)$	Thm:97
172	$\|Au\| \geq c\|u\|$	$\leq$ E(170, 171)
173	Bdd$(A) \wedge$ Injective(A)	$\wedge$I(172)

174	ClosedSet$(R(A)) \wedge$ Surjective(A)	Eq(170, 173)
175	$\exists! u,\ Au = h \wedge h =$ Riesz(f)	$\exists$I(174)
176	$\top$	$\rightarrow$ I(170–175)

Proof of Lemma 113 (Holder)

236	$\|u\|_{L^p} = \|v\|_{L^q} = 1$	Assumption
237	$\|u(x)v(x)\| \leq \frac{\|u(x)\|^p}{p} + \frac{\|v(x)\|^q}{q}$ a.e.	Lem:85
238	$\int_U \|uv\| \leq \frac{1}{p}\|u\|_{L^p}^p + \frac{1}{q}\|v\|_{L^q}^q = 1$	Eq(237)
239	Holder(u, v)	Eq(238, 236)

240	$\top$	$\rightarrow$ I(236–239)

Proof of Lemma 121 (Minkowski)

241	$\|u + v\|_{L^p}^p = \int_U \|u + v\|^p \leq \int_U \|u + v\|^{p-1}(\|u\| + \|v\|)$	$\leq$ E
242	$\int_U \|u + v\|^{p-1}\|u\| \leq \|u + v\|_{L^p}^{p-1}\|u\|_{L^p}$	Thm:108(241)
243	$\int_U \|u + v\|^{p-1}\|v\| \leq \|u + v\|_{L^p}^{p-1}\|v\|_{L^p}$	Thm:108(241)
244	$\|u + v\|_{L^p}^p \leq \|u + v\|_{L^p}^{p-1}(\|u\|_{L^p} + \|v\|_{L^p})$	addMon(242, 243)

245	Minkowski(u, v)	Eq(244)
246	$\top$	$\to$ I(241–245)

Proof of Lemma 122 (GenHolder)

247	$1 \le p_1, \dots, p_m \le \infty \wedge \sum_k p_k^{-1} = 1 \wedge u_k \in L^{p_k}(U)$	Assumption
248	$m = 2 \Rightarrow$ GenHolder(u_k, p_k)	Thm:108
249	IH $\wedge\ q_j := p_j/(1 - p_1^{-1})$	Assumption
250	$\int_U \lvert u_1 \cdots u_m \rvert \le \lVert u_1 \rVert_{L^{p_1}} \lVert u_2 \cdots u_m \rVert_{L^{1/(1-1/p_1)}}$	Thm:108(249)

251	$\lVert u_2 \cdots u_m \rVert_{L^{1/(1-1/p_1)}} \le \prod_{k \ge 2} \lVert u_k \rVert_{L^{p_k}}$	Eq(249)
252	GenHolder(u_k, p_k)	$\le$ E(250, 251)
253	$\top$	$\to$ I(247–252)

Proof of Lemma 123 (Interpolation)

254	$r^{-1} = \theta s^{-1} + (1-\theta) t^{-1} \wedge \text{sum-to-1}(\frac{s}{\theta r}, \frac{t}{(1-\theta)r})$	Def
255	$\int_U \lvert u \rvert^r \le (\int_U \lvert u \rvert^s)^{\theta r/s} (\int_U \lvert u \rvert^t)^{(1-\theta)r/t}$	Thm:108(254)
256	Interpolation(u, r, s, t, θ)	Eq(255)
257	$\top$	$\to$ I(254–256)

Theorems

Proof of Theorem 131 (SupportingHyperplanes)

292	$f : \mathbb{R}^n \to \mathbb{R} \wedge$ Convex$(f) \wedge x \in \mathbb{R}^n$	Assumption
293	epi$(f) := \{(z, t) \mid t \ge f(z)\}$	Def
294	Convex(epi(f))	Def:78(292, 293)
295	$(x, f(x)) \in \partial$epi(f)	Def(293)

296	$\exists (r, -\alpha) \ne 0 : (r, -\alpha) \cdot ((z, t) - (x, f(x))) \le 0$	Lem:91(294, 295)
297	$\alpha \ge 0$	$\le$ E(296)
298	$\alpha > 0$	RAA(297, 296)
299	$r \cdot (z - x) - (t - f(x)) \le 0$	Def(296, 298)

300	$f(z) \ge f(x) + r \cdot (z - x)$	Eq(299)
301	$\forall x\ \exists r\ \forall y,\ f(y) \ge f(x) + r \cdot (y - x)$	$\forall$I(300)
302	$\top$	$\to$ I(292–301)

Proof of Theorem 134 (Jensen)

303	Convex$(f) \wedge$ Open$(U) \wedge$ Bounded$(U) \wedge$ Summable(u)	Assumption
304	$\forall p\ \exists r, \forall q, f(q) \ge f(p) + r \cdot (q - p)$	Thm:126(303)
305	$p :=$ avgE$(\mathrm{u}, U) \wedge q := \mathrm{u}(x) \Rightarrow f(\mathrm{u}(x)) \ge f(p) + r \cdot (\mathrm{u}(x) - p)$	Def(304)
306	$\int_U f(\mathrm{u}) \ge \lvert U \rvert f(p) + r \cdot \left(\int_U \mathrm{u} - \lvert U \rvert p\right) = \lvert U \rvert f(p)$	Def:43(305)

307	$f($avgE$(\mathrm{u}, U)) \le$ avgE$(f \circ \mathrm{u}, U)$	Eq(306)
308	$\top$	$\to$ I(303–307)

Proof of Theorem 106 (GaussGreen)

309	PartitionUnity(ζ_k)	Def
310	$y = \Phi(x) \wedge$ flattens$(\Phi, U, B(x^0, r)) \wedge \det D\Phi = 1$	Def:61(309)
311	$\int_{y_n > 0} \partial_{y_n} (\zeta u)\, dy = \int_{y_n = 0} \zeta u\, dy'$	Lem:33(310)
312	$\int_U (\zeta u)_{x_n}\, dx = \int_{\partial U} \zeta u\, \nu^n\, dS$	Def(311, 310)

313	$\sum_k \zeta_k = 1 \Rightarrow \int_U u_{x_n} = \int_{\partial U} u\nu^n \, dS$	Eq(312)
314	$\int_U u_{x_i} = \int_{\partial U} u\nu^i \, dS$	∀E(313)
315	$\int_U \mathrm{div}(\mathrm{u}) = \int_{\partial U} \mathrm{u} \cdot \nu \, dS$	Eq(314)
316	$\top$	→ I(309–315)

Proof of Theorem 114 (IntParts)

317	$\int_U (uv)_{x_i} = \int_{\partial U} uv\nu^i \, dS$	Thm:101
318	$(uv)_{x_i} = u_{x_i} v + u v_{x_i}$	Lem:35
319	$\mathrm{IntParts}(u, v)$	Eq(317, 318)
320	$\top$	→ I(317–319)

Proof of Theorem 125 (GreenI)

321	$\int_U u_{x_i x_i} = \int_{\partial U} u_{x_i} \nu^i \, dS$	Thm:109
322	$\int_U \mathrm{Lap}(u) = \int_{\partial U} \mathrm{dnu}(u, \nu) \, dS$	Def:60(321)
323	$\top$	→ I(321–322)

Proof of Theorem 126 (GreenII)

324	$\int_U u_{x_i} v_{x_i} = -\int_U u v_{x_i x_i} + \int_{\partial U} u v_{x_i} \nu^i \, dS$	Thm:109
325	$\int_U \mathrm{grad}(v) \cdot \mathrm{grad}(u) = -\int_U u\mathrm{Lap}(v) + \int_{\partial U} u\mathrm{dnu}(v, \nu) \, dS$	Def:24(324)
326	$\top$	→ I(324–325)

Proof of Theorem 133 (GreenIII)

327	$\int_U \mathrm{grad}(u) \cdot \mathrm{grad}(v) = -\int_U v\mathrm{Lap}(u) + \int_{\partial U} v\mathrm{dnu}(u, \nu)$	Thm:121
328	$\mathrm{GreenIII}(u, v)$	Eq(325, 327)
329	$\top$	→ I(327–328)

Proof of Theorem 108 (Polar)

330	$\mathrm{WLOG}(x_0 = 0) \wedge u(x) := \lvert x \rvert \wedge \mathrm{Lipschitz}(u) \wedge \lvert \mathrm{grad}(u) \rvert = 1$ a.e.	Def
331	$\int_{\mathbb{R}^n} f \lvert \mathrm{grad}(u) \rvert = \int_0^\infty \left(\int_{\partial B(0,r)} f \, dS \right) dr$	Thm:102(330)
332	$\int_{\mathbb{R}^n} f = \int_0^\infty \left(\int_{\partial B(0,r)} f \, dS \right) dr$	Eq(331)
333	$\frac{d}{dr} \int_{B(0,r)} f = \int_{\partial B(0,r)} f \, dS$	Lem:33(332)

334	$\top$	→ I(330–333)

Proof of Theorem 107 (Coarea)

335	$\mathrm{Lipschitz}(u) \wedge \mathrm{smooth} - \mathrm{hypersurface}(\{u = r\})$ a.e. r	Assumption
336	$\mathrm{ApproxSimple}(f)$	Assumption
337	$\mathrm{Coarea}(u, f)$	≤ E(335, 336)
338	$\top$	→ I(335–337)

Proof of Theorem 115 (MovingRegions)

339	$X(\tau, \cdot) : U(0) \to U(\tau) \wedge \partial_\tau X = \mathrm{v}$	Def
340	$\int_{U(\tau)} f = \int_{U(0)} f(X(\tau, y), \tau) J(\tau, y) \, dy \wedge J := \det \partial_y X$	Def
341	$\partial_\tau J = J \, \mathrm{div}_x(\mathrm{v})$	Lem:37(340)
342	$\partial_\tau [fJ] = [f_\tau + \mathrm{grad}(f) \cdot \mathrm{v}] J + f J \, \mathrm{div}(\mathrm{v})$	Lem:35(341)

343	$\int_{U(\tau)} [f_\tau + \mathrm{div}(f\mathrm{v})]$	Eq(340, 342)
344	$\int_{U(\tau)} \mathrm{div}(f\mathrm{v}) = \int_{\partial U(\tau)} f\mathrm{v} \cdot \nu \, dS$	Thm:101
345	$\mathrm{MovingRegions}(f, U)$	≤ E(343, 344)
346	$\top$	→ I(339–345)

Proof of Theorem 124 (MollifierProps)

347	$D^\alpha f^\epsilon(x) = \int_U D^\alpha \eta_\epsilon(x-y) f(y)\, dy$	Def:63
348	$f^\epsilon \in C^\infty(U_\epsilon)$	∀I(347)
349	a.e.(avgB($\lvert f - f(x)\rvert, x, r) \to 0$)	Lem:93
350	$\lvert f^\epsilon(x) - f(x)\rvert \le C\,\text{avgB}(\lvert f - f(x)\rvert, x, \epsilon) \to 0$	≤ E(348, 349)

351	$f^\epsilon \to f$ a.e.	∀E(350)
352	$f \in C(U) \Rightarrow \text{modulus}(\omega, f, U)$	Def
353	$\sup_V \lvert f^\epsilon - f\rvert \le C\omega(\epsilon) \to 0$	≤ E(350, 352)
354	$\lVert f^\epsilon\rVert_{L^p(V)} \le \lVert f\rVert_{L^p(W)}$	Thm:108(348)

355	$\lVert f^\epsilon - f\rVert_{L^p(V)} \le 2\lVert f - g\rVert_{L^p(W)} + \lVert g^\epsilon - g\rVert_{L^p(V)}$	≤ E(354)
356	$\text{unif} - \text{on}(g^\epsilon, g, V) \Rightarrow \limsup_{\epsilon\to 0} \lVert f^\epsilon - f\rVert_{L^p(V)} \le 2\delta$	≤ E(355)
357	$\top$	→ I(347–356)

Proof of Theorem 104 (InverseFT)

358	$\text{WLOG}(x_0 = 0) \wedge \mathrm{f}(0) = 0 \wedge Df(0) = I$	Def
359	$\mathrm{f}(x) = x + R(x) \wedge \lVert DR(x)\rVert \le 1/2$ on $\overline{B}(0, r)$	Def(358)
360	$\lvert \mathrm{f}(x) - \mathrm{f}(y)\rvert \ge \frac{1}{2}\lvert x - y\rvert$	≤ E(359)
361	$\forall z \in B(0, r/2)\ \exists x \in B(0, r), \mathrm{f}(x) = z$	Lem:90(359)

362	$D\mathrm{g}(z) = Df(\mathrm{g}(z))^{-1}$	Lem:36(361)
363	$\mathrm{f} \in C^k \Rightarrow \mathrm{g} \in C^k$	∀I(362)
364	$\top$	→ I(358–363)

Proof of Theorem 127 (ImplicitFT)

365	$F(x, y) := (\mathrm{f}(x, y), x) : U \to \mathbb{R}^{m+n}$	Def
366	$DF(x_0, y_0) = \begin{pmatrix} D_x\mathrm{f} & D_y\mathrm{f} \\ I_n & 0 \end{pmatrix} \wedge \det DF = \pm \det D_y\mathrm{f} \neq 0$	Def(365)
367	$\text{LocallyInvertible}(F, (x_0, y_0))$	Thm:99(366)
368	$\mathrm{g}(x) := 2\text{nd} - \text{comp}(F^{-1}(z_0, x))$	Def

369	$\mathrm{f}(x, \mathrm{g}(x)) = z_0 \wedge \mathrm{g}(x_0) = y_0$	∧I(367, 368)
370	$\mathrm{f} \in C^k \Rightarrow \mathrm{g} \in C^k$	∀E(363, 367)
371	$\top$	→ I(365–370)

Proof of Theorem 130 (ArzelaAscoli)

372	$\text{WLOG}(K = \overline{B}(0, R))$	Assumption
373	$\exists\epsilon-\text{net}(\{x_1, \ldots, x_N\}, K)$	∃I(372)
374	$\exists\{k_j\}, \forall i, \text{conv}(f_{k_j}(x_i))$	∃E(373)
375	$\lvert f_{k_j}(x) - f_{k_l}(x)\rvert \le 2\epsilon + \lvert f_{k_j}(x_i) - f_{k_l}(x_i)\rvert$	∀E

376	$\text{unif} - \text{on}(f_{k_j}, f, K) \wedge \text{Continuous}(f)$	≤ E(374, 375)
377	$\top$	→ I(372–376)

Proof of Theorem 105 (ClosedGraph)

378	$G(A) := \{(u, Au)\} \subset X \times Y \wedge \text{ClosedSet}(G(A))$	Def
379	$\text{Banach}(X \times Y) \wedge \lVert(u, v)\rVert = \lVert u\rVert_X + \lVert v\rVert_Y$	Def:76
380	$P : G(A) \to X \wedge (u, Au) \mapsto u \wedge \text{Bijective}(P) \wedge \text{Linear}(P)$	Def(378)
381	$\text{Bdd}(P^{-1})$	Lem:92(380)

382	$\|Au\| \leq \|P^{-1}\|\|u\|$	$\leq$ E(381)
383	Bdd(A)	Def:50(382)
384	$\top$	$\to$ I(378–383)

Proof of Theorem 102 (Riesz)

385	$N(u^*) := \{v \mid \text{pairing}(u^*, v) = 0\} \wedge \text{ClosedSubspace}(N(u^*))$	Def
386	$u^* = 0 \Rightarrow u = 0$	Eq
387	$u^* \neq 0 \Rightarrow \exists w \in \text{Sperp}(N(u^*)), w \neq 0, \|w\| = 1$	Def:48(385)
388	$u := \overline{\text{pairing}(u^*, w)}\, w$	Def

389	$v - \frac{\text{pairing}(u^*,v)}{\text{pairing}(u^*,w)} w \in N(u^*) \Rightarrow (v, w) = \frac{\text{pairing}(u^*,v)}{\text{pairing}(u^*,w)}$	Def:46(387)
390	$\text{pairing}(u^*, v) = (u, v)$	Eq(388, 389)
391	$(u - u', v) = 0\ \forall v \Rightarrow u = u'$	Eq(390)
392	$\top$	$\to$ I(385–391)

Proof of Theorem 112 (WeakCompact)

393	Reflexive(X) $\wedge$ Bdd($\{u_k\}$)	Assumption
394	WeaklyCompact($\overline{B}(0, M), X$)	Lem:94(393)
395	$\exists \{u_{k_j}\} \subset \{u_k\}, u \in X$, WeakConv($u_{k_j}, u$)	$\exists$E(394)
396	$\top$	$\to$ I(393–395)

Proof of Theorem 119 (CompactAdjoint)

397	Bdd($\{u_k\}$) $\Rightarrow \exists \{u_{k_j}\}$, WeakConv($u_{k_j}, u$)	Thm:107
398	$\|K^* u_{k_j} - K^* u\|^2 = (KK^* u_{k_j} - KK^* u, u_{k_j} - u)$	Def:81(397)
399	$KK^* u_{k_j} \to KK^* u$	Def:89(397)
400	$\|K^* u_{k_j} - K^* u\|^2 \to 0$	$\leq$ E(398, 399)

401	$K^* u_{k_j} \to K^* u$	Eq(400)
402	CompactOp(K^*)	Def:89(401)
403	$\top$	$\to$ I(397–402)

Proof of Theorem 128 (Fredholm)

404	$\dim N(I - K) = \infty \Rightarrow \exists \{u_k\} \subset N(I - K) \wedge \text{ONB}(u_k) \wedge \|u_k - u_l\| = \sqrt{2} \wedge Ku_k = u_k$	Def:47
405	$\|Ku_k - Ku_l\| = \sqrt{2} \Rightarrow \neg\text{CompactOp}(K)$	$\bot$I(404)
406	$\exists \gamma > 0, \forall u \in \text{Sperp}(N(I - K)), \|u - Ku\| \geq \gamma \|u\|$	$\forall$E(405)
407	ClosedSet($R(I - K)$)	$\leq$ E(406)

408	$R(I - K) = \text{Sperp}(N(I - K^*))$	Def:48(407)
409	$N(I - K) = \{0\} \Rightarrow R(I - K) = H$	$\forall$I
410	$R(I - K) = H \Rightarrow N(I - K) = \{0\}$	Eq(408, 409)
411	$\dim N(I - K) \geq \dim \text{Sperp}(R(I - K))$	$\leq$ E(409)

412	$\dim N(I - K^*) \geq \dim N(I - K)$	$\forall$E(411)
413	$\dim N(I - K) = \dim N(I - K^*)$	Eq(412)
414	$\top$	$\to$ I(404–413)

Proof of Theorem 135 (SpectrumCompact)

415	$0 \notin \text{sigma}(K) \Rightarrow \text{CompactOp}(I = K \circ K^{-1})$	Def:68
416	$\dim H = \infty \Rightarrow \neg\text{CompactOp}(I)$	$\bot$I(415)
417	$\eta \in \text{sigma}(K) \setminus \{0\} \Rightarrow \text{Eigenvalue}(K, \eta)$	Thm:123(416)
418	$\eta_k \to \eta \neq 0 \Rightarrow \neg\text{CompactOp}(K)$	$\bot$I(417)

419	finite $-$ or $-$ tends0(sigma(K) $\setminus$ {0})	$\forall$E(417, 418)
420	$\top$	$\to$ I(415–419)

Proof of Theorem 136 (SpectralThm)

421	$H_0 = N(S) \wedge H_k = N(S - \eta_k I) \wedge 0 < \dim H_k < \infty$	Thm:123
422	$u \in H_k \wedge v \in H_l \wedge k \neq l \Rightarrow \eta_k(u,v) = \eta_l(u,v) \Rightarrow (u,v) = 0$	Def:88(421)
423	$\widetilde{H} := \widehat{\oplus H_k} \wedge S(\text{Sperp}(\widetilde{H})) \subseteq \text{Sperp}(\widetilde{H})$	Def:48(422)
424	$\widetilde{S} := S\vert_{\text{Sperp}(\widetilde{H})} \wedge \text{CompactOp}(\widetilde{S}) \wedge \text{Symmetric}(\widetilde{S}) \wedge \text{sigma}(\widetilde{S}) = \{0\}$	Def(421)

425	$(\widetilde{S}u, u) = 0\ \forall u \Rightarrow \widetilde{S} = 0$	Lem:98(424)
426	$\text{Sperp}(\widetilde{H}) \subset N(S) \subset \widetilde{H} \Rightarrow \text{Sperp}(\widetilde{H}) = \{0\} \Rightarrow \widetilde{H} = H$	$\forall$E(425)
427	ONB $-$ of $-$ eigenvectors(S)	$\forall$I(421, 426)
428	$\top$	$\to$ I(421–427)

Proof of Theorem 117 (LebesgueExistence)

429	$\text{Box}(I) :\Leftrightarrow I = \Pi_{i=1}^n (a_i, b_i] \wedge \text{vol}(I) := \Pi_{i=1}^n (b_i - a_i)$	Def
430	$\forall A \subseteq \mathbb{R}^n$, $\text{outmeas}(A) := \inf\{\Sigma_k \text{ vol}(I_k) \mid A \subseteq \cup_k I_k \wedge \text{Box}(I_k)\}$	Def
431	Monotone(outmeas) $\wedge$ countably $-$ subadditive(outmeas) $\wedge$ outmeas($\emptyset$) $= 0$	$\leq$ E(430)
432	$\mathcal{M} := \{A \subseteq \mathbb{R}^n \mid \forall E, \text{outmeas}(E) = \text{outmeas}(E \cap A) + \text{outmeas}(E \setminus A)\}$	Def

433	SigmaAlgebra($\mathcal{M}$)	Def:72(432)
434	$\vert\cdot\vert := \text{outmeas}\vert_{\mathcal{M}} \wedge$ countably $-$ additive($\vert\cdot\vert$)	Def(432, 433)
435	$\forall I$, Box(I) $\Rightarrow \vert I\vert = \text{vol}(I) \Rightarrow \vert B\vert = \text{vol}(B)$	$\forall$I(430)
436	Open(U) $\Rightarrow U \in \mathcal{M}$	$\forall$E(432)

437	Closed(F) $\Rightarrow F \in \mathcal{M}$	$\forall$E(433, 436)
438	Null $-$ subset : $\vert A\vert = 0 \wedge B \subseteq A \Rightarrow B \in \mathcal{M} \wedge \vert B\vert = 0$	$\forall$E(432, 431)
439	$\exists \mathcal{M}, \vert\cdot\vert$, SigmaAlgebra($\mathcal{M}$) $\wedge$ Open(U) $\wedge$ Closed(U) $\subset \mathcal{M} \wedge \vert B\vert = \text{vol}(B) \wedge$ countably $-$ additive($\vert\cdot\vert$)	$\exists$I(433, 434, 435, 436, 437, 438)
440	$\top$	$\to$ I(429–439)

Proof of Theorem 118 (Egoroff)

441	$A_j^m := \cap_{k \geq j}\{\vert f_k - f\vert < 1/m\}$	Def
442	$A_j^m \uparrow\ \bar{A}^m \wedge \bar{A}^m := \{\lim f_k = f, \vert f - f\vert \leq 1/m\} \wedge \vert A \setminus \bar{A}^m\vert = 0$	$\leq$ E(441)
443	$\vert A \setminus A_j^m\vert \to 0$	$\leq$ E(442)
444	$\exists j(m), \vert A \setminus A_{j(m)}^m\vert < \epsilon/2^m$	$\exists$I(443)

445	$E := \cap_m A_{j(m)}^m \wedge$ Measurable(E)	Def:73
446	$\vert A \setminus E\vert \leq \epsilon \wedge$ unif $-$ on(f_k, f, E)	$\leq$ E(444)
447	$\top$	$\to$ I(441–446)

Proof of Theorem 110 (Fatou)

448	$g_k := \inf_{j \geq k} f_j \wedge$ Measurable(g_k) $\wedge$ Nondecreasing(g_k)	Def:73
449	$\int g_k \uparrow\ \int \liminf f_k$	Thm:104(448)
450	$g_k \leq f_j\ \forall j \geq k \Rightarrow \int g_k \leq \inf_{j \geq k} \int f_j$	$\leq$ E(448)
451	$\int \liminf f_k \leq \liminf \int f_k$	$\leq$ E(449, 450)

452	$\top$	$\to$ I(448–451)

Proof of Theorem 109 (MonotoneConv)

453	$f := \sup_k f_k = \lim_k f_k \wedge \int f_k \uparrow \; L$	Def
454	$\int f_k \le \int f \Rightarrow L \le \int f$	$\le$ E(453)
455	$0 \le s \le f \wedge \text{Simple}(s) \wedge E_k := \{f_k \ge (1-\epsilon)s\} \uparrow \; \mathbb{R}^n$	Def:44
456	$(1-\epsilon)\int_{E_k} s \le \int f_k \le L \Rightarrow \int s \le L$	$\le$ E(455, 453)

457	$\int f \le L$	$\forall$E(456)
458	$\int \lim f_k = \lim \int f_k$	$\le$ E(454, 457)
459	$\top$	$\to$ I(453–458)

Proof of Theorem 111 (DominatedConv)

460	$\int (g+f) \le \liminf \int (g+f_k)$	Thm:105
461	$\int f \le \liminf \int f_k$	$\le$ E(460)
462	$\int (g-f) \le \liminf \int (g-f_k)$	Thm:105
463	$\limsup \int f_k \le \int f$	$\le$ E(462)

464	$\int f \le \liminf \int f_k \le \limsup \int f_k \le \int f$	$\le$ E(461, 463)
465	$\int f_k \to \int f$	Eq(464)
466	$\top$	$\to$ I(460–465)

Proof of Theorem 116 (LebesgueDiff)

467	LocSummable$(f) \wedge u(x) :=$ avgB$(\lvert f - f(x)\rvert, x, r)$	Def:41
468	subharmonic $-$ like$(u) \wedge \int_{B(x,r)} u \to 0$	Lem:93(467)
469	avgB$(\lvert f - f(x_0)\rvert, x_0, r) \to 0$ a.e. x_0	$\forall$E(468)
470	avgB$(f, x_0, r) \to f(x_0)$ a.e.	$\le$ E(469)

471	$\top$	$\to$ I(467–470)

Proof of Theorem 120 (Pettis)

472	StronglyMeasurable(f) $\Rightarrow$ WeaklyMeasurable(f)	Def:65
473	StronglyMeasurable(f) $\Rightarrow$ AlmostSeparablyValued(f)	$\forall$E(472)
474	WeaklyMeasurable(f) $\wedge$ AlmostSeparablyValued(f) $\Rightarrow$ StronglyMeasurable(f)	$\wedge$I(472, 473)
475	$\top$	$\to$ I(472–474)

Proof of Theorem 129 (Bochner)

476	$\lVert s_k\rVert \le \lVert s_k - f\rVert + \lVert f\rVert \wedge \int \lVert f\rVert = \lim \int \lVert s_k\rVert < \infty$	Def:44
477	$\exists\{s_k\} \wedge$ Simple$(s_k) \wedge s_k \to f \wedge N_k := \{\lVert s_k - f\rVert > 1\} \wedge s'_k := s_k \cdot 1_{[0,T]\setminus N_k}$	$\exists$I(476)
478	$\int \lVert s'_k - f\rVert \le \int \min(1, \lVert s_k - f\rVert) \to 0$	Thm:106(477)
479	$\lVert \int s'_k\rVert \le \int \lVert s'_k\rVert$	$\le$ E

480	$\lVert \int f\rVert \le \int \lVert f\rVert$	$\le$ E(478, 479)
481	pairing$(u^*, \int f) = \lim \int$ pairing$(u^*, s'_k) = \int$ pairing(u^*, f)	Def:53(478)
482	$\top$	$\to$ I(476–481)